Group 1 · Fernão de Oliveira · Statement of Work: your studies, drawings and specializations

Your studies and drawings

Your group's list from the Statement of Work: 32 items, 5 of them drawings. Tick each one off as it is done.

ID	Title	Deliverable	Status
STU-D1.1	Owner's requirements and port limits	D1	
STU-D1.2	Environmental targets	D1	
STU-D2.1	Reference ship database with sources	D2	
STU-D2.2	Regressions and bounds	D2	
STU-D3.1	Ship model	D3	
STU-D3.2	Closed weight balance	D3	
STU-D4.1	Voyage model	D4	
STU-D4.2	Financial model	D4	
STU-D5.1	Validation against reference ships	D5	
STU-D6.1	Optimization	D6	
STU-D7.1	EEDI and CII	D7	
STU-D7.2	Freeboard and intact stability	D7	
STU-D7.3	Draught at every port	D7	
STU-D8.1	Hydrostatics check	D8	
DWG-D8.1	Profile (A3)	D8	
DWG-D8.2	Midship section (A3)	D8	
STU-D10.1	Datasheet (template 0.4)	D10	
STU-D10.2	Presentation, 15 minutes	D10	
STU-B.1	Capacity count on own arrangement	D8	
DWG-B.1	Container and battery layout with crane reach	D8	
STU-SP01.1	Lightship and deadweight distribution	D9 · SP01	
STU-SP01.2	Still-water and wave bending moments and shear forces	D9 · SP01	
STU-SP01.3	Allowable stresses, bending and shear	D9 · SP01	
STU-SP01.4	Section modulus, moment of inertia and neutral axis	D9 · SP01	
STU-SP01.5	Allowable envelopes of vertical bending moment and shear force	D9 · SP01	
STU-SP01.6	Four critical loading conditions	D9 · SP01	
STU-SP01.7	Preliminary scantlings (calculation report, spreadsheet)	D9 · SP01	
DWG-SP01.1	Midship cross-section	D9 · SP01	
STU-SP02.1	Optimum B-series propeller	D9 · SP02	
STU-SP02.2	Engine matching	D9 · SP02	

ID	Title	Deliverable	Status
STU-SP02.3	PTO/PTI or electric operating modes	D9 · SP02	
DWG-SP02.1	Stern arrangement	D9 · SP02	

Your specializations

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

- STU-SP01.1 lightship weight and deadweight distribution along the length.
- STU-SP01.2 still-water and wave bending moments and shear forces.
- STU-SP01.3 allowable stresses, in bending and in shear.
- STU-SP01.4 section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.5 allowable envelopes of the vertical bending moment and of the shear force: the total allowable moment is the section modulus times the allowable bending stress, $M_{total} = Z \cdot \sigma_{allow}$, and the allowable still-water moment is what the wave moment leaves, $M_{still} = M_{total} - M_{wave}$; the same for the shear force.
- STU-SP01.6 four critical loading conditions (subject to justification and/or definition with the instructor), each checked against the envelopes.
- STU-SP01.7 preliminary scantlings of the midship cross-section: a separate calculation report, as a spreadsheet.
- DWG-SP01.1 midship cross-section.

SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

- STU-SP02.1 optimum Wageningen B-series propeller for the service speed.
- STU-SP02.2 engine matching: operating points on the motor layout diagram, with the sea margin.
- STU-SP02.3 the two-shaft electric modes (Baseline B).
- DWG-SP02.1 stern arrangement: propeller, shaft line and rudder.

The full text of every item is in the Statement of Work.